

Advanced Optimization

Chapter 18 — Sequential Quadratic Programming

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 18

SQP is Newton's method wearing a QP costume

Equality problem $\min f$ s.t. $c(x) = 0$. Its KKT conditions are a square nonlinear system:

$$F(x, \lambda) = \begin{bmatrix} \nabla f(x) - A(x)^T \lambda \\ c(x) \end{bmatrix} = 0 \quad (18.3)$$

— so run **Chapter 11's Algorithm 11.1** on it. The Newton system is

$$\begin{bmatrix} \nabla_{xx}^2 \mathcal{L}_k & -A_k^T \\ A_k & 0 \end{bmatrix} \begin{bmatrix} p_k \\ p_\lambda \end{bmatrix} = \begin{bmatrix} -\nabla f_k + A_k^T \lambda_k \\ -c_k \end{bmatrix}, \quad (18.6)$$

well-defined under **Assumptions 18.1**: LICQ + $\nabla_{xx}^2 \mathcal{L}$ positive definite on $\text{Null}(A)$ — Ch. 16's Lemma 16.1 conditions, at work again.

The second reading (18.7): a quadratic program

$$\min_p f_k + \nabla f_k^T p + \frac{1}{2} p^T \nabla_{xx}^2 \mathcal{L}_k p \quad \text{s.t. } A_k p + c_k = 0$$

has the *same* solution: rearranging (18.6) gives (18.9) with $\lambda_{k+1} = l_k$, the QP multiplier. Quadratic model of the *Lagrangian*, linear model of the constraints.

Algorithm 18.1: solve (18.7), set $(x_{k+1}, \lambda_{k+1}) = (x_k + p_k, l_k)$. Newton-on- F powers the *analysis* (rates free from Thm 11.2); the QP reading powers the *algorithms* — a QP, unlike a linear system, survives inequalities.

Inequalities: the QP inherits them, and identifies the active set

For the general problem, linearize everything (IQP subproblem):

$$\min_p f_k + \nabla f_k^T p + \frac{1}{2} p^T \nabla_{xx}^2 \mathcal{L}_k p \quad \text{s.t.} \quad \nabla c_i^T p + c_i = 0 \ (i \in \mathcal{E}), \quad \nabla c_i^T p + c_i \geq 0 \ (i \in \mathcal{I}). \quad (18.11)$$

Chapter 16 solves it; the QP's active set is the guess of $\mathcal{A}(x^*)$.

Theorem 18.1 (Robinson): the guess locks in

Under LICQ + strict complementarity + SOSC at (x^*, λ^*) : for (x_k, λ_k) close enough, the subproblem (18.11) has a local solution whose active set **equals** $\mathcal{A}(x^*)$. From then on SQP behaves as Newton on the correct equality-constrained problem — **quadratic endgame**, combinatorics finished.

Two architectures for the active-set search

IQP: solve the full inequality QP each iteration — expensive per step, but Ch. 16's warm-start row now pays out: near convergence, the previous active set is (by Thm 18.1) almost always right, so the QP costs almost nothing.

EQP: pick a working set first (cheaply), then solve one *equality* QP — SLQP (§18.5) uses an LP to pick it; gradient projection (§18.6) uses a projected path.

Between Algorithm 18.1 and a real solver: remote starts, indefinite $\nabla_{xx}^2 \mathcal{L}_k$, infeasible linearizations, a step-acceptance rule — §18.3–18.5.

Ingredient 1: when the linearization is infeasible — go elastic

The IQP subproblem can be **infeasible** even when the NLP is fine: constraints $x \leq 1$, $x^2 \geq 4$ linearized at $x_k = 1$ give $-p \geq 0$ and $2p - 3 \geq 0$ — **no p exists**.

Elastic mode (18.12) = Chapter 17's ℓ_1 reformulation

$$\min_{x, v, w, t} f(x) + \mu \sum_{\mathcal{E}} (v_i + w_i) + \mu \sum_{\mathcal{I}} t_i \quad \text{s.t. } c_i(x) = v_i - w_i, \quad c_i(x) \geq -t_i, \quad v, w, t \geq 0.$$

Its linearization is *always* feasible (slacks absorb anything); for μ beyond the Ch. 17 threshold, solutions coincide with the NLP's. This is precisely SNOPT's **elastic mode** — and if the NLP itself is infeasible, the method gracefully converges to a stationary point of the infeasibility measure instead.

Trust-region SQP has its own infeasibility problem ($\text{radius} \cap \text{linearization} = \emptyset$) with its own three answers — §18.5.

Ingredient 2: Hessians — Chapter 6 pays its last debt

Quasi-Newton on the Lagrangian: update B_k with

$$s_k = x_{k+1} - x_k, \quad y_k = \nabla_x \mathcal{L}(x_{k+1}, \lambda_{k+1}) - \nabla_x \mathcal{L}(x_k, \lambda_{k+1}). \quad (18.13)$$

New trouble: $\nabla_{xx}^2 \mathcal{L}$ can be indefinite *even at the solution* (only the reduced Hessian must be PD — Ch. 12), so $s_k^T y_k > 0$ is no longer guaranteed by any line search. Skipping updates when (18.14) fails is crude and can stall.

Procedure 18.2: damped BFGS (the fix promised in Ch. 6)

Replace y_k by the interpolant $r_k = \theta_k y_k + (1 - \theta_k) B_k s_k$ with

$$\theta_k = 1 \text{ if } s_k^T y_k \geq 0.2 s_k^T B_k s_k; \quad \theta_k = \frac{0.8 s_k^T B_k s_k}{s_k^T B_k s_k - s_k^T y_k} \text{ otherwise,} \quad (18.15)$$

then BFGS with (s_k, r_k) . Guarantee: $s_k^T r_k \geq 0.2 s_k^T B_k s_k > 0$ — positive definiteness survives (Ch. 6's Wolfe argument, transplanted). $\theta = 1$: pure BFGS; $\theta = 0$: keep B_k ; in between: pull toward the old matrix just enough.

Damped BFGS works well but papers over the real issue (the Hessian *is* indefinite): **SR1** + **trust region** is the honest pairing — SR1's habitat, third sighting. Large scale: limited-memory versions (Ch. 7), which is what SNOPT and KNITRO actually run.

Ingredient 2b: reduced-Hessian updating

Look at the step equations (18.9) through Ch. 16's null-space lens ($p = Yp_Y + Zp_Z$):

$$(A_k Y_k) p_Y = -c_k, \quad (Z_k^T \nabla_{xx}^2 \mathcal{L}_k Z_k) p_Z = -Z_k^T \nabla_{xx}^2 \mathcal{L}_k Y_k p_Y - Z_k^T \nabla f_k. \quad (18.19)$$

- ▶ The range-space part p_Y is fixed by feasibility alone — **the Hessian only ever acts through $Z_k^T \nabla_{xx}^2 \mathcal{L}_k Z_k$** . So approximate just that: maintain $M_k \approx Z_k^T \nabla_{xx}^2 \mathcal{L}_k Z_k$, an $(n-m) \times (n-m)$ matrix — near the solution it *is* positive definite (Thm 12.6), so plain BFGS applies without damping.
- ▶ Cheaper cross-term variants drop $Z^T \nabla_{xx}^2 \mathcal{L} Y p_Y$ (it vanishes at the rate the constraints are satisfied); multipliers from the QP (18.20) or least-squares estimates $\hat{\lambda} = (AA^T)^{-1} A \nabla f$ (18.21).
- ▶ The price: everything lives in coordinates of Z_k , which must vary smoothly with k for the theory (and the updates) to make sense — and superlinearity weakens to *two-step* superlinear (Thm 18.7). No free lunch, but a much smaller lunch bill.

Which to use? Full (damped, limited-memory) QN when $n - m$ is large or Z is awkward; reduced when $n - m \ll n$ — e.g. discretized control and network problems.

Ingredient 3: the ℓ_1 merit function, and how much μ it needs

Accept $\alpha_k p_k$ only if it decreases $\phi_1(x; \mu) = f(x) + \mu \|c(x)\|_1$ sufficiently (Armijo on the directional derivative, (18.28)) — Ch. 17's exact penalty, now employed as judge rather than as method.

Theorem 18.2 (the SQP step is a ϕ_1 -descent direction)

For p_k from (18.9):

$$D(\phi_1(x_k; \mu); p_k) = \nabla f_k^T p_k - \mu \|c_k\|_1 \leq -p_k^T \nabla_{xx}^2 \mathcal{L}_k p_k - (\mu - \|\lambda_{k+1}\|_\infty) \|c_k\|_1. \quad (18.29\text{--}18.30)$$

Proof shape: Taylor + $A_k p_k = -c_k$ makes the ℓ_1 term contract *linearly* ($\|c_k + \alpha A_k p_k\|_1 = (1 - \alpha) \|c_k\|_1$); the multiplier bound comes from substituting the first KKT row.

- ▶ Descent needs $\mu > \|\lambda_{k+1}\|_\infty$ (18.32) and (reduced) positive definiteness — Ch. 17's exactness threshold, live on stage.
- ▶ Chasing $\|\lambda\|_\infty$ directly misbehaves; the robust rule asks the *model* for a certificate: pick μ so the predicted merit decrease is at least $\rho\mu\times$ the predicted feasibility decrease, $q_\mu(0) - q_\mu(p_k) \geq \rho\mu[m(0) - m(p_k)]$ (18.35) — one inequality serving line search and trust region alike.

Ingredient 4: the Maratos effect, and the second-order correction

Example 18.1 (= Example 15.4): the merit function rejects a perfect step

On $\min -\cos\theta$ -type problem (15.34), the pure SQP step from a point *on* the unit circle moves tangentially: f and $\|c\|$ **both increase** at the full step even though it is exactly the Newton step toward x^* . ϕ_1 vetoes it, α_k collapses, and quadratic convergence dies — the **Maratos effect**. The merit function is honest but myopic: the quadratic loop of the circle is invisible to its linear model.

Second-order correction (SOC)

After a rejected p_k , evaluate $c(x_k + p_k)$ and re-solve the QP with shifted constraints

$$\nabla c_i^T p + d_i = 0, \quad d_i = c_i(x_k + p_k) - \nabla c_i^T p_k \quad (18.42)$$

— d_i is a free estimate of the *curvature* of c_i along p_k (Taylor (18.40) with $p \approx p_k$ (18.41)). The corrected step hugs the constraint surface; near a SOSC solution, full steps get accepted and the fast rate returns.

Cost: one extra constraint evaluation — so trigger SOC only when the rejection comes with rising $\|c\|$. Alternative cure: a **filter** (Ch. 15, skipped): keep a Pareto list of $(\|c\|, f)$ pairs, accept any step not dominated by history — no μ at all. FILTERSQP is built on it.

Algorithm 18.3: the line search SQP method, fully assembled

Algorithm 18.3, annotated with its supply chain

Step	Where it came from
solve IQP (18.11) for $p_k, \hat{\lambda}$ $p_\lambda = \hat{\lambda} - \lambda_k$	Ch. 16 Alg 16.3, warm-started from $\mathcal{W}_{k-1}$ dual step toward the QP multiplier
choose μ_k by (18.36), $\sigma = 1$	model certificate (18.35)
backtrack α_k on ϕ_1 Armijo	Ch. 3 + Thm 18.2 descent
$(x, \lambda) += \alpha_k(p_k, p_\lambda)$	
update B_k : damped BFGS on (18.13)	Ch. 6 + Procedure 18.2

- ▶ Omitted for clarity, needed in production: elastic mode for infeasible QPs, SOC for Maratos, filter as an alternative judge. Add those and **you have sketched SNOPT**.
- ▶ Exact-Hessian variant: modify $\nabla_{xx}^2 \mathcal{L}_k$ during factorization until the reduced part is PD — the “one object, many names” regularization again.
- ▶ Everything large-scale: limited-memory BFGS + the Ch. 16 QP machinery.

Byrd–Omojokun (Algorithm 18.4): split the step, reuse every tool

Adding $\|p\| \leq \Delta_k$ to (18.7) can make the subproblem **infeasible** (the linearized constraints may not intersect the ball). Relax the constraint to $A_k p + c_k = r_k$ and choose r_k as the *best achievable* within $0.8\Delta_k$:

Two subproblems per iteration

Normal step (feasibility): $v_k = \arg \min \|A_k v + c_k\|_2^2$ s.t. $\|v\| \leq 0.8\Delta_k$ — solved by the **dogleg** (Ch. 4) with $p^B = -A_k^T(A_k A_k^T)^{-1}c_k$; set $r_k = A_k v_k + c_k$.

Tangential step (optimality): minimize the quadratic model s.t. $A_k p = r_k$, $\|p\| \leq \Delta_k$ — solved by **projected CG (Alg 16.2) with Steihaug exits** (Ch. 7), started from v_k .

- ▶ Judge: the (unsquared) ℓ_2 merit $\phi_2 = f + \mu\|c\|_2$ with ratio ρ_k (18.47–18.48), Ch. 4 radius logic, μ by (18.35). Indefinite $\nabla_{xx}^2 \mathcal{L}_k$ is *welcome* (Steihaug); SR1 fits naturally.
- ▶ Cost: one AA^T /augmented factorization + Hessian-vector products. **This skeleton, moved inside a barrier method, is exactly KNITRO/CG — Ch. 19's Algorithm 19.4.**

$S\ell_1$ QP and SLQP: penalty and LP variants

$S\ell_1$ QP (Fletcher)

Put the linearized constraints *into the objective* as ℓ_1 terms (Ch. 17's model (17.30)) and keep only $\|p\|_\infty \leq \Delta_k$: an always-feasible **elastic QP** (18.49–18.50). Merit = ϕ_1 itself — model and merit finally agree, so no Maratos drama; box TR fits LP/QP machinery.

SLQP (the EQP flagship; KNITRO/ACTIVE)

Split the work: **(1)** drop the quadratic term, solve the ℓ_1 -penalized *LP* in an ℓ_∞ box (18.53) $\rightarrow$ active set of the LP becomes the working set $\mathcal{W}_k$ + Cauchy step; **(2)** solve the *equality* QP on $\mathcal{W}_k$ (quadratic term restored, spherical TR) by projected CG. LP technology picks the set; Ch. 16 technology takes the step; two radii, Δ_k^{LP} tuned to track $\|p_k\|$.

Algorithm 18.5: pricing μ by reference solve

If the QP already achieves linearized feasibility ($m_k(p) = 0$): keep μ . Otherwise solve once with “ $\mu = \infty$ ” (pure feasibility) to learn the *best possible* decrease $m_k(0) - m_k(p_\infty)$, then raise μ until the actual step captures a fixed fraction of it (+ certificate (18.35)). A few extra QPs buy a principled dial — the Ch. 17 update problem, finally given a good answer.

Nonlinear gradient projection: Ch. 16.7 goes nonlinear

For $\min f$ s.t. $l \leq x \leq u$: build the quadratic model q_k with $B_k \succ 0$ (18.59), solve $\min_{l \leq x \leq u} q_k$ *approximately* by **Algorithm 16.5** (Cauchy point on the projected path + CG on the free variables), set $p_k = \hat{x} - x_k$, Armijo backtrack on f .

- ▶ Descent in three lines: $f_k = q_k(x_k) > q_k(x^c) \geq q_k(\hat{x}) = f_k + \nabla f_k^T p_k + \frac{1}{2} p_k^T B_k p_k \Rightarrow \nabla f_k^T p_k < 0$.
- ▶ Trust-region version: intersect the box with $\|x - x_k\|_\infty \leq \Delta_k$ — still a box, $\max(l, x_k - \Delta e) \leq x \leq \min(u, x_k + \Delta e)$, so the same machinery runs; B_k may then be indefinite (exact Hessian or SR1).
- ▶ This EQP-flavored method is the engine of **L-BFGS-B** (limited-memory B_k , bound constraints) and of LANCELOT's subproblem solver (Ch. 17) — the projected path does the active-set combinatorics wholesale, as in Ch. 16.7.

The quiet moral of §18.5–18.6: every practical SQP variant is a *routing* of the same five tools — dogleg, projected CG, Steihaug, ℓ_1 models, active sets — assembled from Chapters 4, 5, 7, 16, 17.

Convergence: global by penalty bookkeeping, fast by Chapter 11

Theorem 18.3 (global, classical)

Line search SQP with PD models B_k , ℓ_1 merit, $\mu \geq \|\lambda_k\|_\infty + \rho$ fixed, iterates and QP solutions bounded in a region where $f, c \in C^1$: every limit point of $\{x_k\}$ is a KKT point. (Assumptions do real work — boundedness excludes degenerating Jacobians; modern analyses relax them.)

Rates: the bill comes to Chapter 11

Thm 18.4: exact Hessian $\Rightarrow (x_k, \lambda_k) \rightarrow (x^*, \lambda^*)$ **quadratically** — proof: SQP is Newton on $F = 0$; apply Thm 11.2. Done.

Thm 18.5 (Dennis–Moré, constrained): quasi-Newton is superlinear iff

$$\frac{\|P_k(B_k - \nabla_{xx}^2 \mathcal{L}_*)(x_{k+1} - x_k)\|}{\|x_{k+1} - x_k\|} \rightarrow 0, \quad P_k = Z_k Z_k^T \quad (18.63)$$

— Ch. 3's Thm 3.6 with a *projection*: only the null-space action of the error matters.

Thm 18.6: BFGS with $\nabla^2 \mathcal{L}_* \succ 0$ and good starts $\Rightarrow$ (18.63) holds: Q-superlinear. Damped BFGS: R-superlinear. Reduced-Hessian ($B_k = Z_k M_k Z_k^T$): only the two-sided projection is controlled $\Rightarrow$ **two-step** superlinear (Thm 18.7).

The rate hierarchy is a precise accounting of *which part of the Hessian you paid for*: all of it, its null-space action, or its two-sided projection.

Perspectives: when SQP is the right tool, and who implements it

- ▶ **SQP shines when:** the active set is large ($n - m$ small — reduced problems are tiny); function evaluations are expensive (SQP is frugal with them, vs. augmented Lagrangian); problems are badly scaled (more robust than interior-point); or a *sequence* of related NLPs must be solved — warm starts, the same advantage as Ch. 16, doubled.
- ▶ **Interior-point (Ch. 19) shines when:** problems are huge and the active set is a mystery — it never touches the combinatorics. The IQP-vs-EQP question within SQP is likewise unsettled.

Package	Type	Globalization	Signature ideas
SNOPT	line search IQP	ℓ_1 merit	elastic mode (18.12), LM-BFGS
FILTERSQP	trust region IQP	filter	feasibility restoration
KNITRO/ACTIVE	SLQP (EQP)	ℓ_1 penalty	LP working set, Alg 18.5, proj. CG
CONOPT	reduced gradient / SQP	—	GRG heritage

Quasi-Newton verdict for constrained problems: BFGS is weaker here than in Ch. 6 (the underlying matrix is often not PD), damped/limited-memory versions are “adequate”; SR1 options can win. The full-Hessian question is genuinely open — a rare honest sentence to end a chapter on.

Chapter 18 in one table

Object	Statement	Role
(18.6) $\equiv$ (18.7)	Newton-on-KKT = QP subproblem	the two readings
Thm 18.1 (Robinson)	QP identifies $\mathcal{A}(x^*)$ locally	quadratic endgame
Elastic mode (18.12)	ℓ_1 relaxation, always feasible	SNOPT; Ch. 17 reused
Procedure 18.2	damped BFGS, $s^T r \geq 0.2s^T B s$	PD survives; Ch. 6's debt paid
Reduced QN (18.19)	update $M_k \approx Z^T \nabla_{xx}^2 \mathcal{L} Z$	small, naturally PD
Thm 18.2 + (18.35)	ϕ_1 descent; model-priced μ	the judge and its salary
SOC (18.42)	constraint-curvature refit	Maratos cured
Alg 18.4 (B–O)	normal dogleg + tangential proj. CG	KNITRO/CG; $\rightarrow$ Ch. 19
SLQP + Alg 18.5	LP picks set; reference-solve μ	KNITRO/ACTIVE
Thms 18.4–18.7	quadratic / superlinear / 2-step	rate = Hessian part paid for

Two ideas to carry forward

SQP is Newton's method on the KKT system, read as a QP so that inequalities survive the reading — everything else is globalization bookkeeping. The merit function is a judge, not an oracle: it can veto Newton's best steps (Maratos), and the SOC/filter machinery exists to overrule it.

Homework (1/2)

Derivations

- 1 Show the equivalence of the Newton system (18.6) and the QP (18.7): derive (18.9) and identify $\lambda_{k+1} = l_k$.
- 2 For damped BFGS (Procedure 18.2), prove $s_k^T r_k \geq 0.2 s_k^T B_k s_k$ for both branches of (18.15), and conclude positive definiteness of B_{k+1} via Ch. 6's argument.
- 3 Fill in the proof of Theorem 18.2: derive both Taylor bounds and the substitution that yields (18.30).
- 4 Work Example 18.1: derive the SQP step $p_k = (\sin^2 \theta, -\sin \theta \cos \theta)^T$ (18.38), and verify that at the full step both f and $|c|$ increase — the Maratos effect in two lines of trigonometry.
- 5 Derive the SOC constraint shift (18.42) from the Taylor argument (18.39)–(18.41), and explain why d_i costs one constraint evaluation and no derivatives.
- 6 Compare (18.63) with Theorem 3.6's Dennis–Moré condition: what exactly does the projection P_k excuse the update from learning, and which theorem (18.6 or 18.7) exploits that excuse?

Homework (2/2)

Implementation

7. *Local SQP*. Implement Algorithm 18.1 for equality constraints and run it on the circle problem (17.3) from several starts. Verify quadratic convergence of $\|(x_k, \lambda_k) - (x^*, \lambda^*)\|$ and exhibit one remote start where it diverges — the case for globalization.
8. *Line search SQP*. Assemble Algorithm 18.3 using your Ch. 16 active-set QP solver, the (18.36) rule for μ , and damped BFGS. Solve Example 16.4's QP-turned-NLP (perturb the objective to be non-quadratic) and a 5–10 variable test problem; log warm-start QP iteration counts along the run — watch them fall to near zero as Thm 18.1 kicks in.
9. *Maratos experiment*. On problem (15.34)/Example 18.1, run item-8's method with and without SOC from x_0 near (but not on) the circle. Plot α_k and $\|x_k - x^*\|$: without SOC, unit steps get rejected and the rate degrades; with SOC, the fast rate returns.
10. *Optional, lab-flavored*. Write a 2-bus AC-OPF (one generator, one load, quadratic cost, voltage bounds) and solve it with item 8's SQP. Compare iterates and the final active set against IPOPT's. Two solvers, two philosophies — the comparison is the point; Ch. 19 dissects the other one.

Next: Ch. 19 — *interior-point methods for nonlinear programming* (two weeks; destination: IPOPT dissected).